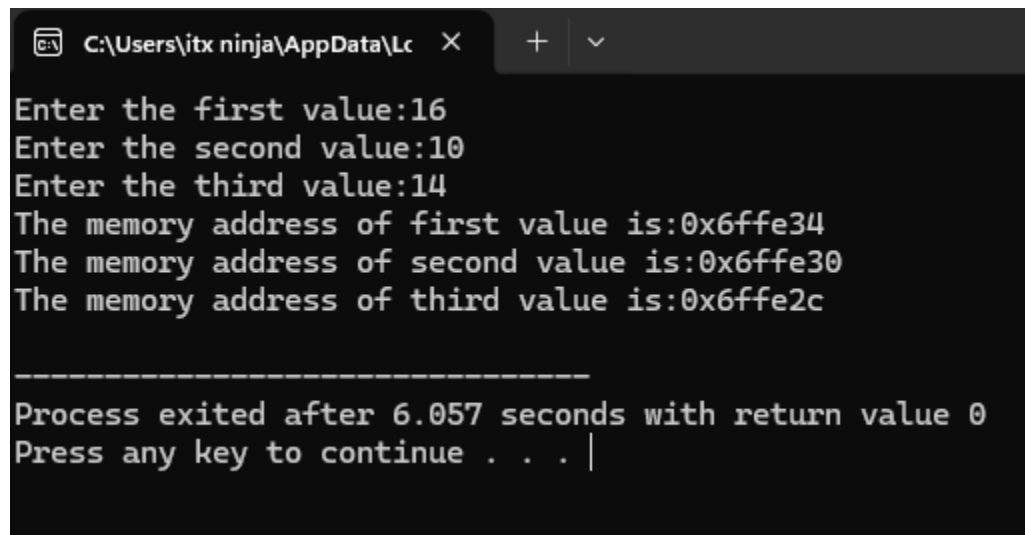


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Lab task

TASK 1: Write a C++ program that prints the addresses of three variables using reference Operator.

```
task1]
#include<iostream>
using namespace std;
int main()
{
    int num1=0;
    int num2=0;
    int num3=0;
    int *ptr1;
    ptr1=&num1;
    int *ptr2;
    ptr2=&num2;
    int *ptr3;
    ptr3=&num3;
    cout<<"Enter the first value:";
    cin>>num1;
    cout<<"Enter the second value:";
    cin>>num2;
    cout<<"Enter the third value:";
    cin>>num3;
    cout<<"The memory address of first value is:"<<ptr1<<endl;
    cout<<"The memory address of second value is:"<<ptr2<<endl;
    cout<<"The memory address of third value is:"<<ptr3<<endl;
    return 0;
}
```



The screenshot shows a terminal window with the following output:

```
C:\Users\itx ninja\AppData\Local\Microsoft\Windows\Terminal\... X + v
Enter the first value:16
Enter the second value:10
Enter the third value:14
The memory address of first value is:0x6ffe34
The memory address of second value is:0x6ffe30
The memory address of third value is:0x6ffe2c
-----
Process exited after 6.057 seconds with return value 0
Press any key to continue . . . |
```

TASK 2: Write a program that inputs four integer values and passes them to a function using pointers. The function then exchanges the value of 1st integer with 2nd integer and 3rd with 4th integer and then the main program finally displays the values.

```
LAB TASK 2:
#include<iostream>
using namespace std;
void pointers(int *ptr1, int *ptr2, int *ptr3, int *ptr4)
{
    cout<<"The first value before swapping is:"<<*ptr1<<endl;
    cout<<"The second value before swapping is:"<<*ptr2<<endl;
    cout<<"The third value before swapping is:"<<*ptr3<<endl;
    cout<<"The fourth value before swapping is:"<<*ptr4<<endl;
    int n=0;
    n=*ptr1;
    *ptr1=*ptr2;
    *ptr2=n;
    n=*ptr3;
    *ptr3=*ptr4;
    *ptr4=n;
    cout<<"The first value after swapping is:"<<*ptr1<<endl;
    cout<<"The second value after swapping is:"<<*ptr2<<endl;
    cout<<"The third value after swapping is:"<<*ptr3<<endl;
    cout<<"The forth value after swapping is:"<<*ptr4<<endl;
}
int main()
{
    int num1=0;
    int num2=0;
    int num3=0;
    int num4=0;
    int *ptr1;
    int *ptr2;
    int *ptr3;
    int *ptr4;
    cout<<"Enter the first value:";
    cin>>num1;
    cout<<endl;
    cout<<"Enter the second value:";
```

```
C:\Users\itx ninja\AppData\Local\Microsoft\Windows\Terminal\... X + v
Enter the first value:15
Enter the second value:19
Enter the third value:18
Enter the fourth value:17
The first value before swapping is:15
The second value before swapping is:19
The third value before swapping is:18
The fourth value before swapping is:17
The first value after swapping is:19
The second value after swapping is:15
The third value after swapping is:17
The forth value after swapping is:18
-----
Process exited after 14.49 seconds with return value 0
Press any key to continue . . . |
```

TASK 3: Write a program to input arrays values by using functions and pointer array. Make a function of name “inputArray” in which user enters the value of array and then display the values in main program.

```

LAB TASK 3:
#include<iostream>
using namespace std;
void inputarray(int array[5])
{
    int *ptr;
    ptr=array;
    for(int i=0; i<5; i++)
    {
        cout<<"Enter the "<<i+1<<" value:";
        cin>>*ptr;
        ptr++;
    }
}
int main()
{
    int *ptr;
    int array[5];
    inputarray(array);
    ptr=array;
    for(int i=0; i<5; i++)
    {
        cout<<"The"<<i+1<<" value using pointer are"<<*ptr<<endl;
        ptr++;
    }
    return 0;
}

```

```

C:\Users\itx ninja\AppData\Local... X + v
Enter the 1 value:1
Enter the 2 value:44
Enter the 3 value:56
Enter the 4 value:76
Enter the 5 value:89
The1 value using pointer are1
The2 value using pointer are44
The3 value using pointer are56
The4 value using pointer are76
The5 value using pointer are89

-----
Process exited after 14.59 seconds with return value 0
Press any key to continue . . . |

```

TASK 4: create a menu-based program that consists of the following features using functions and pointer:
Find the largest of three numbers.

Find the smallest of three numbers.

Find an average of three numbers.

```
task4
#include<iostream>
using namespace std;
void comparison_function(int user,int *num1,int *num2,int *num3)
{
    int average=0;

    if(user==1)
    {
        if(*num1>*num2 && *num1>*num3)
        {
            cout<<"The number 1 is greatest."<<endl;
        }
        else if(*num2>*num1 && *num2>*num3)
        {
            cout<<"The number 2 is greatest."<<endl;
        }
        else if (*num3>*num1 && *num3>*num2)
        {
            cout<<"The number 3 is greatest."<<endl;
        }
        else
            cout<<"You entered the wrong number."<<endl;
    }
    if(user==2)
    {
        if(*num1<*num2 && *num1<*num3)
        {
            cout<<"The number 1 is smallest."<<endl;
        }
        else if(*num2<*num1 && *num2<*num3)
        {
            cout<<"The number 2 is smallest."<<endl;
        }
        else if(*num3<*num1 && *num3<*num2)
        {
            cout<<"The number 3 is smallest."<<endl;
        }
    }
}
```

```
C:\Users\itx ninja\AppData\Local\Microsoft\Windows\Terminal\... X + v
Enter the first value:99
Enter the second value:89
Enter the third value:79
1. Find Largest number.
2. Find the Smallest number.
3. Find average.
4. End.
Enter the number according to menu to perform the desired task:1
The number 1 is greatest.
1. Find Largest number.
2. Find the Smallest number.
3. Find average.
4. End.
Enter the number according to menu to perform the desired task:89
1. Find Largest number.
2. Find the Smallest number.
3. Find average.
4. End.
Enter the number according to menu to perform the desired task:|
```

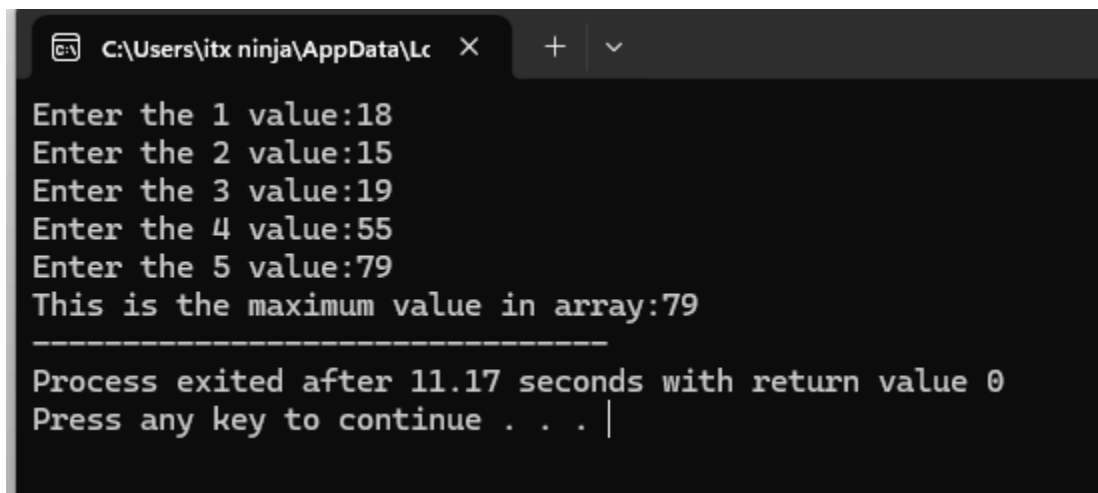
Home task

Write a function `findMax(int* ptr, int size)` that takes a pointer to an array and its size, then returns the maximum value. Call this function from main and print the result.

Objective: Write logic to find the largest element in an array using only pointers.

Home TASK 1

```
#include<iostream>
using namespace std;
void findMax(int *ptr,int size)
{
    int max=0;
    for(int i=0; i<5; i++)
    {
        cout<<"Enter the "<<i+1<<" value:";
        cin>>*ptr;
        ptr++;
    }
    ptr=ptr-size;
    max=*ptr;
    for(int i=0; i<5; i++)
    {
        if(*ptr>max)
        {
            max=*ptr;
        }
        ptr++;
    }
    cout<<"This is the maximum value in array:"<<max;
}
int main()
{
    int size=5;
    int array[5];
    int *ptr1;
    ptr1=array;
    findMax(array,5);
    return 0;
}
```



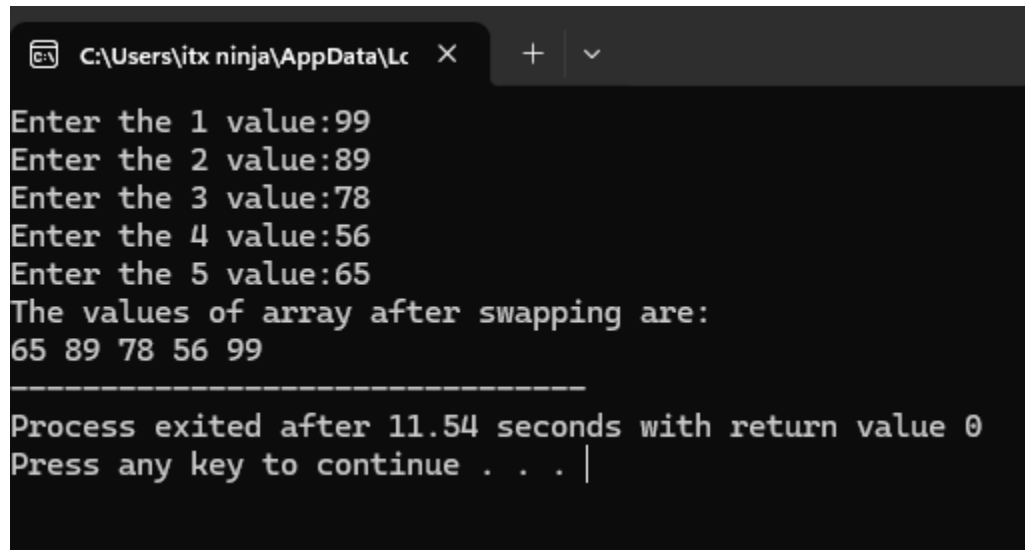
```
C:\Users\itx ninja\AppData\Local... X + v
Enter the 1 value:18
Enter the 2 value:15
Enter the 3 value:19
Enter the 4 value:55
Enter the 5 value:79
This is the maximum value in array:79
-----
Process exited after 11.17 seconds with return value 0
Press any key to continue . . . |
```

Task2

```

#include<iostream>
using namespace std;
void swapping(int *ptr,int size)
{
    int temp;
    for(int i=0; i<5; i++)
    {
        cout<<"Enter the "<<i+1<<" value:";
        cin>>*(ptr+i);
    }
    int *ptr1=ptr;
    int *ptr2=ptr+size-1;
    while(ptr<ptr2)
    {
        temp=*ptr1;
        *ptr1=*ptr2;
        *ptr2=temp;
        ptr1++;
        ptr2--;
    }
    cout<<"The values of array after swapping are:"<<endl;
    for(int i=0; i<5; i++)
    {
        cout<<*(ptr+i)<<" ";
    }
}
int main()
{
    int array[5];

```



```

C:\Users\itx ninja\AppData\Local
Enter the 1 value:99
Enter the 2 value:89
Enter the 3 value:78
Enter the 4 value:56
Enter the 5 value:65
The values of array after swapping are:
65 89 78 56 99
-----
Process exited after 11.54 seconds with return value 0
Press any key to continue . . . |

```

Task3

```

}*/
/*HOME TASK 3:
#include<iostream>
using namespace std;
void vowels(char array[25])
{
    cin.getline(array,25);
    int count=0;
    char *ptr=array;
    while(*ptr!='\0')
    {
        if(*ptr=='a' || *ptr=='A' || *ptr=='E' || *ptr=='e' || *ptr=='I' || *ptr=='i' || *ptr=='O' || *ptr=='o' || *ptr=='U' || *ptr=='u')
        {
            count++;
        }
        ptr++;
    }
    cout<<count;
}
int main()
{
    char array[25];
    vowels(array);
    return 0;
}*/

```

```

C:\Users\itx ninja\AppData\Local
55
0
-----
Process exited after 9.544 seconds with return value 0
Press any key to continue . . . |

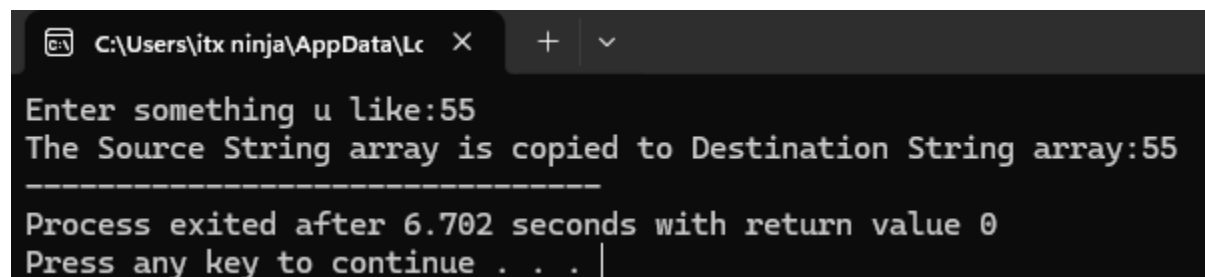
```

Task4

```

/*HOME TASK 4:
#include<iostream>
using namespace std;
void myStringCopy(char *dest, char *src)
{
    while(*src!='\0')
    {
        *dest=*src;
        src++;
        dest++;
    }
    *dest='\0';
}
int main()
{
    char src[25];
    char dest[25];
    cout<<"Enter something u like:";
    cin.getline(src,25);
    myStringCopy(dest, src);
    cout<<"The Source String array is copied to Destination String array:";
    char *temp=dest;
    while(*temp!='\0')
    {
        cout<<*temp;
        temp++;
    }
    return 0;
}*/

```



```

C:\Users\itx ninja\AppData\Local...
Enter something u like:55
The Source String array is copied to Destination String array:55
-----
Process exited after 6.702 seconds with return value 0
Press any key to continue . . . |

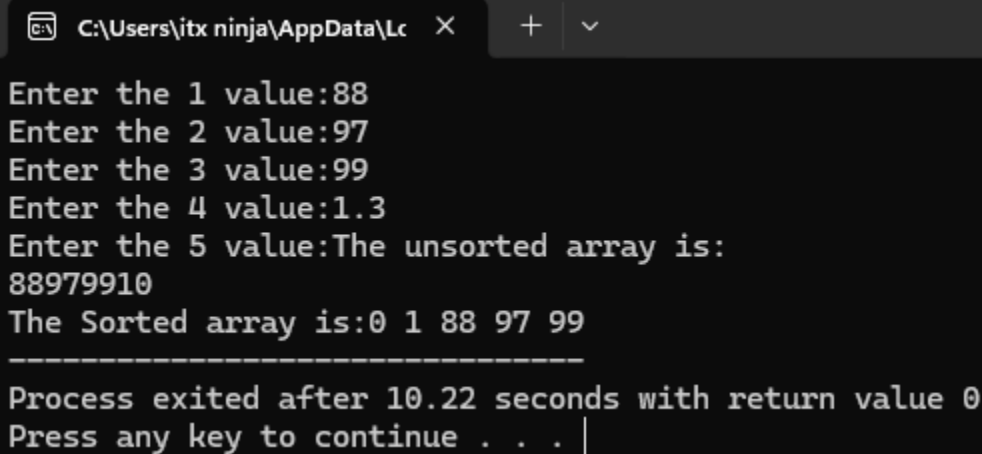
```

Task5

```

#include<iostream>
using namespace std;
void bubbleSort(int *arr, int n)
{
    int temp;
    for(int i=0; i<=n-2; i++)
    {
        for(int j=0; j<=n-i-2; j++)
        {
            if(*(arr+j)>*(arr+j+1))
            {
                temp=*(arr+j);
                *(arr+j)=*(arr+j+1);
                *(arr+j+1)=temp;
            }
        }
    }
    cout<<"The Sorted array is:";
    for(int i=0; i<5; i++)
    {
        cout<<*(arr+i)<<" ";
    }
}
int main()
{
    int n=5;
    int values[5];
    int *arr;
    arr=values;
    for(int i=0; i<5; i++)
    {
        cout<<"Enter the "<<i+1<<" value:";
        cin>>*(arr+i);
    }
    cout<<"The unsorted array is:"<<endl;
    for(int i=0; i<5; i++)

```



```

C:\Users\itx ninja\AppData\Local
Enter the 1 value:88
Enter the 2 value:97
Enter the 3 value:99
Enter the 4 value:1.3
Enter the 5 value:10
The unsorted array is:
88979910
The Sorted array is:0 1 88 97 99
-----
Process exited after 10.22 seconds with return value 0
Press any key to continue . . . |

```

Task6

```

/*task6
#include<iostream>
using namespace std;

int myStrLen(char *s)
{
    int count = 0;

    while(*(s + count) != '\0')
    {
        count++;
    }

    return count;
}

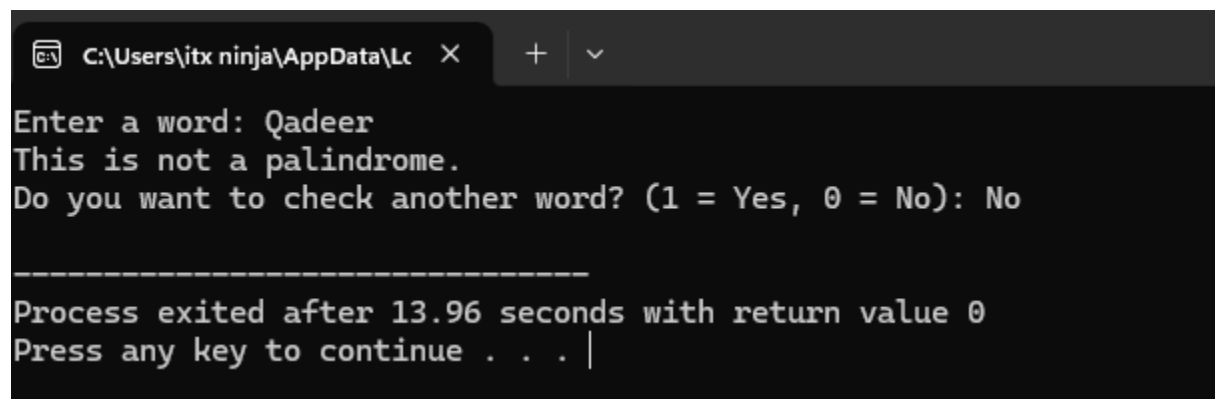
bool isPalindrome(char *s)
{
    int len = myStrLen(s);

    char *start = s;
    char *end = s + len - 1;

    while(start < end)
    {
        if(*start != *end)
        {
            return false;
        }
        start++;
        end--;
    }

    return true;
}

```



```

C:\Users\itx ninja\AppData\Local... X + v
Enter a word: Qadeer
This is not a palindrome.
Do you want to check another word? (1 = Yes, 0 = No): No

-----
Process exited after 13.96 seconds with return value 0
Press any key to continue . . . |

```

Task7

```

/*task7
#include<iostream>
using namespace std;
int main()
{
    int rows, cols;
    cout<<"Enter number of rows: ";
    cin>>rows;
    cout<<"Enter number of columns: ";
    cin>>cols;
    int **matrix = new int*[rows];
    for(int i=0; i<rows; i++)
    {
        matrix[i] = new int[cols];
    }
    for(int i=0; i<rows; i++)
    {
        for(int j=0; j<cols; j++)
        {
            matrix[i][j] = i * cols + j + 1;
        }
    }
    cout<<"\nMatrix is:\n";

    for(int i=0; i<rows; i++)
    {
        for(int j=0; j<cols; j++)
        {
            cout<<matrix[i][j]<<"\t";
        }
        cout<<endl;
    }
    for(int i=0; i<rows; i++)
    {
        delete[] matrix[i];
    }
}

```

C:\Users\itx ninja\AppData\Local

Enter number of rows: 5
Enter number of columns: 3

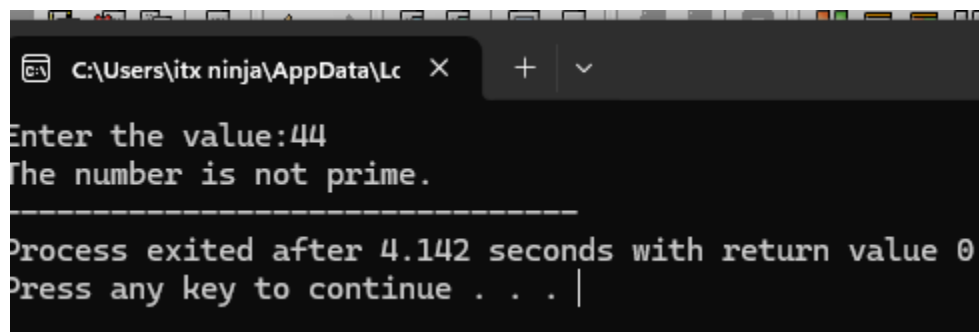
Matrix is:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Process exited after 5.749 seconds with return value 0
Press any key to continue . . . |

Task8

```
/*HOME TASK 8:
#include<iostream>
using namespace std;
void primeNumber(int *ptr)
{
    int count=0;
    for(int i=2; i<*ptr-1; i++)
    {
        if(*ptr%i==0)
        {
            count++;
        }
    }
    if(count==0)
    {
        cout<<"The number is prime.";
    }
    else
        cout<<"The number is not prime.";
}
int main()
{
    int num=0;
    cout<<"Enter the value:";
    cin>>num;
    int *ptr=&num;
    primeNumber(&num);
    return 0;
}
1*/
```



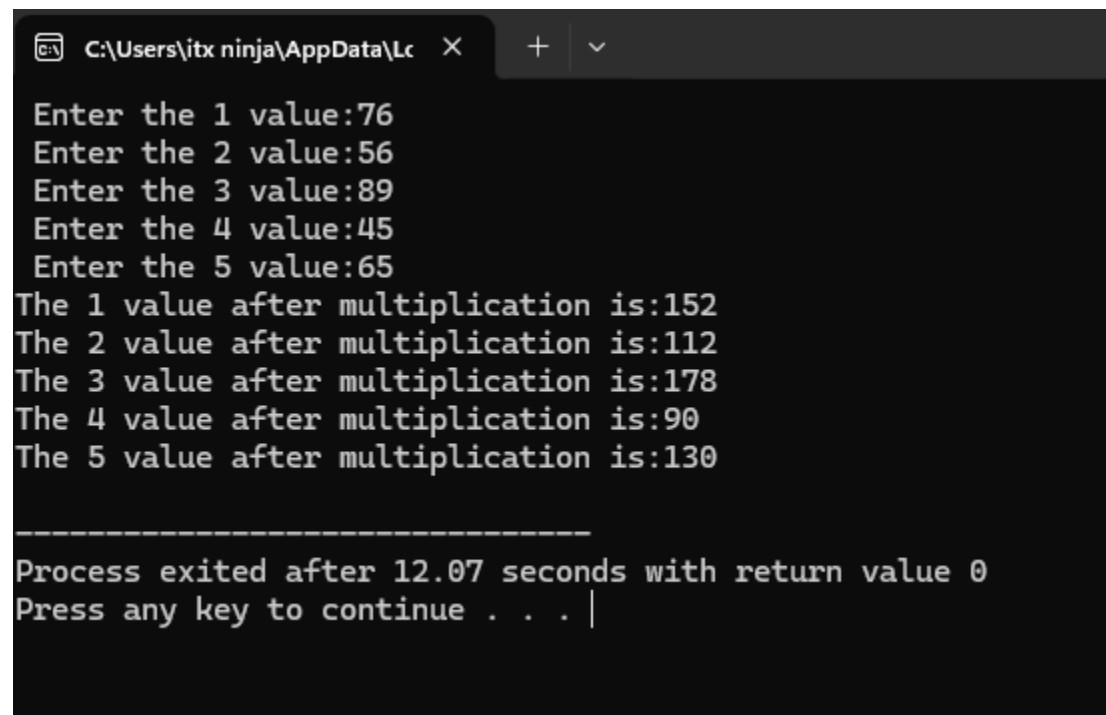
```
C:\Users\itx ninja\AppData\Local\Microsoft\Windows\Terminal\... X + v
Enter the value:44
The number is not prime.
-----
Process exited after 4.142 seconds with return value 0
Press any key to continue . . . |
```

Task9

```

/*HOME TASK 9:
#include<iostream>
using namespace std;
void inputArray(int *ptr)
{
    for(int i=0; i<5; i++)
    {
        cout<<" Enter the "<<i+1<<" value:";
        cin>>*ptr;
        ptr++;
    }
}
void doubArray(int *ptr)
{
    int *ptr2=ptr;
    for(int i=0; i<5; i++)
    {
        *ptr2=(*ptr2)*2;
        ptr2++;
    }
    ptr2=ptr;
    for(int i=0; i<5; i++)
    {
        cout<<"The "<<i+1<<" value after multiplication is:"<<*ptr2<<endl;
        ptr2++;
    }
}
}

```



```

C:\Users\jtx ninja\AppData\Local... X + v
Enter the 1 value:76
Enter the 2 value:56
Enter the 3 value:89
Enter the 4 value:45
Enter the 5 value:65
The 1 value after multiplication is:152
The 2 value after multiplication is:112
The 3 value after multiplication is:178
The 4 value after multiplication is:90
The 5 value after multiplication is:130

-----
Process exited after 12.07 seconds with return value 0
Press any key to continue . . . |

```